

New Operative Surgical Concept of Two Fascial Layers Enveloping the Recurrent Laryngeal Nerve

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ABSTRACT

Background. Avoiding recurrent laryngeal nerve (RLN) injury is aided by a precise anatomical understanding of the RLN before entry into the larynx where it is at greatest risk.

Methods. A literature review and study of 977 RLNs at operation have been used to describe the last 2 cm of the anatomical course of the extralaryngeal RLN.

Results. The tertiary branches of the inferior thyroid artery pass in a superficial vascular fascial layer anterolateral to the RLN. The tubercle of Zuckerkandl and the superior parathyroid glands lie within this superficial vascular fascial covering layer. The RLN lies on a deeper layer of more fibrous tissue, the ligament of Berry. In all 977 RLNs, the RLN was sandwiched between an anterolateral vascular layer and a medial layer, the ligament of Berry.

Conclusions. The site of greatest risk during thyroidectomy to the RLN is in the 2-cm course of the extralaryngeal nerve above the trunk of the inferior thyroid artery where tension forms an artificial genu of the nerve. Once the superficial vascular fascial and Berry's ligament layers are released, the nerve falls back into the tracheoesophageal groove and adopts a serpiginous appearance, indicating completeness of the dissection of the lobe from the lateral trachea and division of the ligament of Berry.

Thyroidectomy is a common operation, and the most important complication is recurrent laryngeal nerve (RLN) palsy.¹ RLN palsy occurs in up to 10% of cases on a temporary basis, but it is permanent in 0.3% to 2% of cases.^{2,3} Many clinical scenarios are associated with a potential increased risk of RLN palsy, including thyroid

cancer, reoperative surgery, retrosternal goiter, large goiters, bifid RLNs, non-recurrent laryngeal nerves, where the RLN is anterior to the inferior thyroid artery, Graves' disease, thyroiditis, postoperative hemorrhage, low-volume hospital and surgeon surgery, and failure to expose the full extent of the RLN.

The precise pathophysiological cause of recurrent nerve palsy in most cases is unknown, but it is presumed to be associated with stretching and compression at fixed point sites, which in turn causes neurapraxia, which resolves in most cases. Rare causes of injury include division, entrapment within a ligature, thermal injury, electrical injury, ischemia, crushing with a hemostat, and pharmacological and viral causes.

However, regardless of the cause of the RLN palsy, a permanent unilateral palsy is important because it will cause dysphonia, weakened cough, predisposition to aspiration, and social and economic detrimental effects to the patient, as well as medicolegal implications. A bilateral recurrent nerve palsy is potentially life-threatening and may necessitate a permanent tracheostomy, although this is rare.

One of the key aims of thyroid surgery is to minimize as much as possible the incidence of RLN palsy. A prerequisite to this is a thorough understanding of the clinical associations, the presumed causes, and a precise knowledge of the surgical anatomy and surgical technique required to safely remove the thyroid without injuring the RLN.

There is a considerable volume of literature indicating that the site of greatest risk of injury to the RLN during thyroidectomy is the portion of the RLN in its course above the inferior thyroid artery in the region of the ligament of Berry, before entry into the larynx.^{4,5} In this region, which is usually approximately the last 2 cm of the extralaryngeal course of the RLN, the nerve is often <1 mm away from the thyroid and is attached by relatively inflexible fibrous tissues to the ligament of Berry. It is also potentially the site of the greatest effect of traction on the nerve.

Much has been written about the relationship between the trunk of the inferior thyroid artery before division, and the RLN and whether the nerve is anterior or posterior to the trunk. Also of importance is the relationship between the terminal tertiary branches of the inferior thyroid artery and the RLN above the level of the trunk of the inferior thyroid artery.

The last 2 cm of the extralaryngeal course of the RLN is a small regional area of anatomy of critical importance in thyroidectomy because it incorporates the relationship of the RLN, and whether it is nonbifid or bifid, with the ligament of Berry, the tubercle of Zuckerkandl, and the superior parathyroid gland, all in a compact area in which precise surgical technique and understanding of anatomy is critical to successful thyroidectomy (Fig. 1). This area of anatomy is often briefly, incompletely, and inconsistently described in a range of anatomical, general surgical, endocrine surgery, and head and neck texts.^{6–14}

Accordingly, here, I describe in detail the last 2 cm of the extralaryngeal course of the RLN and highlight the important anatomical considerations that allow safe and total removal of the thyroid lobe. In addition, I present a new operative surgery anatomical concept of two fascial layers surrounding the final course of the extralaryngeal RLN, and I attempt to define precisely what constitutes the ligament of Berry. Throughout, although there are many

variations in thyroid anatomy, only the common and important variants are described.

PATIENTS AND METHODS

A literature review was undertaken to clarify the description of the exact anatomy of the ligament of Berry and the final 2-cm course of the extralaryngeal RLN. Over the last 6 years, I studied the RLN at thyroidectomy, recording branches and the relation of the nerve to the ligament of Berry. I studied a total of 663 thyroidectomies (314 total thyroidectomy and 349 hemithyroidectomy) in patients of mean age 53 (range, 12–93) years, with a female-to-male ratio of 4:1, resulting in 977 nerves. The overall thyroid pathology was benign in 612 (92.3%), predominately multinodular goiter, and malignant in 51 (7.7%).

All patient details were entered into the Monash University Endocrine Surgery Unit Thyroidectomy Database. Human Research and Ethics Committee approval was granted for collection and analysis of the data. This enabled a new description of this area of anatomy and the presentation of a new concept of two fascial layers surrounding the RLN.

RESULTS AND DISCUSSION

Comparison of In Situ Anatomy to Operative Anatomy

The in situ normal anatomy will differ to that seen at operation because it is disturbed by the pathology of the thyroid gland and because surgery rotates and delivers the thyroid lobe anteromedially through almost 120°. The posterior and lateral aspects of the thyroid lobe mold itself to adjacent structures in the normal in situ position (Fig. 2).

When performing a hemithyroidectomy, the first step after dividing the middle thyroid vein is to enter the loose areolar tissue plane between the common carotid artery and

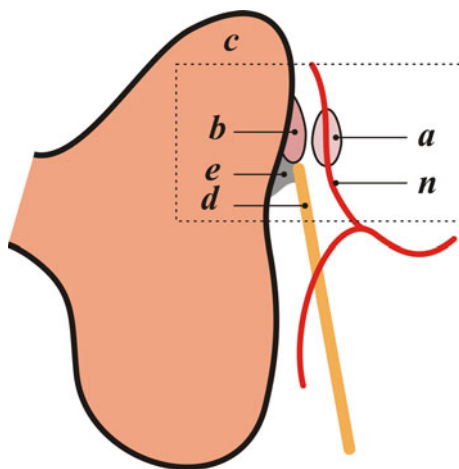


FIG. 1 The region of the final 2 cm extralaryngeal course of the recurrent laryngeal nerve (RLN). Left thyroid lobe from in front. Parts: a, superior parathyroid gland; b, tubercle Zuckerkandl; c, superior pole of thyroid gland; d, RLN; e, ligament of Berry; f, thyroid gland; g, common carotid artery; h, internal jugular vein; i, trachea; j, esophagus; k, sternohyoid muscle; l, sternothyroid muscle; m, sternocleidomastoid muscle; n, inferior thyroid artery; o, cricoid cartilage; p, superior thyroid vessels; q, fascia attaching posterior “v” lip of superior pole; r, posterior “v” lip; s, approximate level of entry RLN beneath cricopharyngeus; t, superficial vascular fascial layer containing branches inferior thyroid artery overlying the RLN; u, lower border cricopharyngeus; v, loose fascia attaching RLN to ligament of Berry; 1, surface 1; 2, surface 2; 3, surface 3

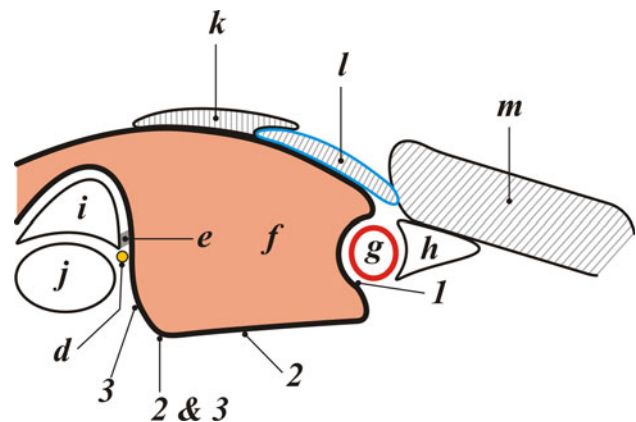


FIG. 2 Transverse section of thyroid lobe in situ. Parts as in Fig. 1

the lateral surface of the thyroid, designated surface 1 in Fig. 2. This then allows access to mobilize the thyroid lobe forward through a further 90°, exposing the posterior surface 2. At the junction between surfaces 2 and 3, the tertiary branches of the inferior thyroid artery and tributaries of the vein enter and leave the thyroid lobe. Progressive mobilization and forward rotation of the thyroid lobe expose this. The result is rotation of the thyroid lobe through approximately 120°, which in turn pulls the RLN forward, applying a tensile force to it and straightening it (Fig. 3). It also tends to pull the recurrent nerve forward out of the tracheoesophageal groove, forming an artificial genu above the level of the inferior thyroid artery, which is the site of greatest risk to the RLN.¹⁵ Surface 3 in Fig. 2 is exposed only when the tertiary branches of the inferior thyroid vein and artery are divided, thereby exposing the RLN.

Similarities and Differences of Right and Left RLNs Above and Below the Inferior Thyroid Artery

Below the level of the inferior thyroid artery, there are marked differences between the right and left RLNs. On the left side, the recurrent nerve recurs around the arch of the aorta at the level of the ligamentum arteriosum. The left RLN therefore tends to reach the tracheoesophageal groove low down in the neck and ascend within the groove up toward the larynx. Further, the left RLN is longer than the right RLN. On the right side, the RLN recurs around the first part of the subclavian artery, and at a level below the inferior thyroid artery, it tends to be more anterolateral to the tracheoesophageal groove than the left RLN. As a consequence, the right RLN is shorter than the left RLN. Upon reaching the level of the inferior thyroid artery, the nerve passes either behind, in front of, or between the

branches of the inferior thyroid artery at the level of its trunk (Figs. 4 and 5).^{16,17} Ardito et al., in a study of >2500 RLNs, found the RLN anterior to the right inferior thyroid artery in 12%, and the left in 1.9%, posterior to the right inferior thyroid artery in 61% and the left in 77.4%, and between the branches on the right in 27% and the left in 20.5%.¹⁶ Steinberg et al. reported the RLN between the branches of the inferior thyroid artery in 6.5%, posterior to the inferior thyroid artery in 61.5%, and anterior to the artery in 32.5%.¹⁸ The more usual anatomy where the RLN passes behind the trunk of the inferior thyroid artery is predominately described below.

Above the level of the inferior thyroid artery trunk, however, both right and left RLNs progress in a similar course essentially in the tracheoesophageal groove, with minimal differences between the two sides. Whether the RLN is front of or behind the trunk of the inferior thyroid artery does not alter the need to formally identify and preserve the RLN. Of potentially greater importance is the relationship between the tertiary branches of the inferior thyroid artery and the RLN above the level of the inferior thyroid artery, which tends to be more constant.

Posterior “V” Lip of the Superior Thyroid Pole

Before ligating the superior thyroid pole vessels, the superior pole itself must be mobilized. This involves careful dissection medially in the cricothyroid space to preserve the external laryngeal nerve and also posterior mobilization. Posteriorly in section, the superior pole forms an inverted “v” lip that is attached by fascia posteriorly. This fascia must be divided to free the posterior “v” lip of the superior pole.

The superior thyroid vessels and the posterior “v” lip of the superior thyroid pole in addition provide an anatomical key and direction to the layer of fascia containing the point of entry of the tertiary branches of the inferior thyroid vein

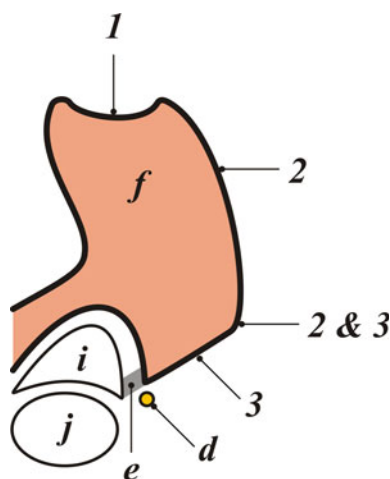


FIG. 3 Anatomy thyroid lobe after 120° anteromedial rotation. Parts as in Fig. 1

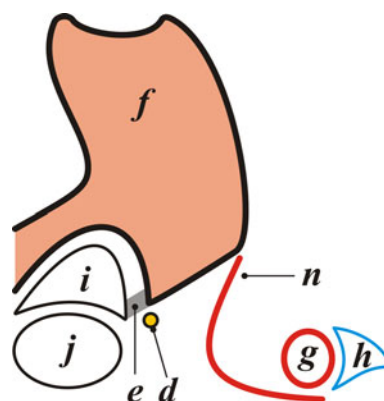


FIG. 4 Recurrent laryngeal nerve posterior to trunk of inferior thyroid artery, transverse section. Parts as in Fig. 1

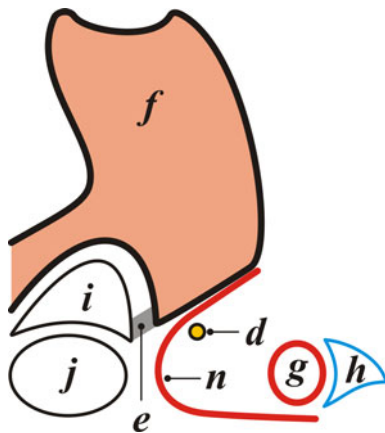


FIG. 5 Recurrent laryngeal nerve anterior to trunk of inferior thyroid artery, transverse section. Parts as in Fig. 1

and artery into the thyroid lobe, which lies lateral to the RLN at a level above the inferior thyroid artery trunk (in the situation where the RLN is posterior to the trunk of the inferior thyroid artery).

The superior pole with its posterior “v” lip is seen in transverse section in Fig. 6. This is at a level above the cricothyroid joint and cricopharyngeus, below which the RLN enters the larynx. This layer of fascia from the posterior “v” lip, passing posteriorly, requires division before the superior thyroid pole can be mobilized.¹⁹

The posterior “v” lip, however, continues down along the posterior aspect of the thyroid gland slightly lateral to the RLN, between surfaces 2 and 3 in Fig. 2, which is the point of entry of the inferior thyroid vessels into the lobe. When the thyroid lobe is rotated 120° anteromedially, this fascial layer containing the vessels becomes anterolateral to the RLN at a level above the inferior thyroid artery trunk. This I have termed the superficial vascular fascial layer.

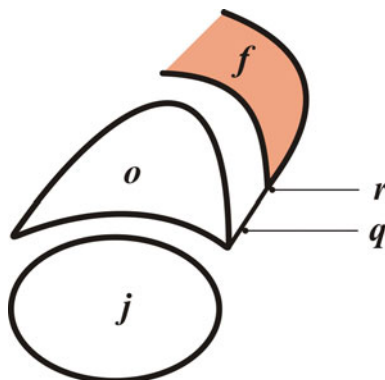


FIG. 6 Superior pole thyroid, demonstrating posterior “v” lip, transverse section. Parts as in Fig. 1

Superficial Vascular Fascial Layer Overlying the RLN Above the Level of the Inferior Thyroid Artery

The superficial fascial layer described above is vascular and overlies the RLN. Therefore, when looking down onto the lateral and posterior aspect of the thyroid lobe, the RLN will not be visible above the level of the inferior thyroid artery at this stage of the operation. Figure 7 shows the superior pole vessels and indicates the approximate level of entry of the RLN into the larynx under the cricopharyngeus, behind the cricothyroid joint. Below this level, above the trunk of the inferior thyroid artery, a fascial layer is present containing the tertiary branches and tributaries of the inferior thyroid vein and artery, which covers the RLN by this vascular fascial layer, regardless of whether the RLN is posterior or anterior to the trunk of the inferior thyroid artery.

Tubercle of Zuckerkandl

Within this superficial vascular layer is also located the tubercle of Zuckerkandl. Embryologically, this derives from the fourth pharyngeal pouch and represents the site of fusion of the ultimobranchial body that brings to the thyroid, in association with the superior parathyroid gland, the parafollicular cells, which subsequently produce

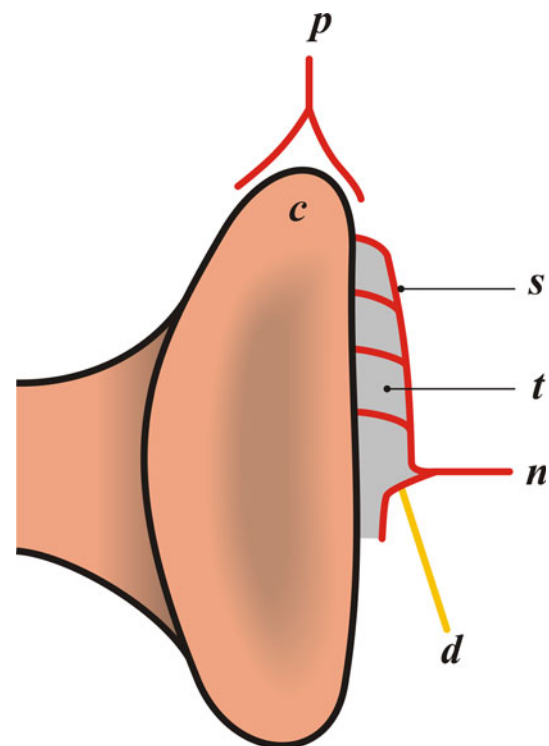


FIG. 7 Left thyroid lobe, rotated medially, viewed from anterior, surface 2 shaded, showing the superficial vascular layer covering the recurrent laryngeal nerve above the level of the inferior thyroid artery. Parts as in Fig. 1

calcitonin.²⁰ The tubercle of Zuckerkindl is therefore a useful anatomical pointer to the superior parathyroid gland, which tends to be located above the inferior thyroid artery within the superficial vascular fascial layer. The tubercle theoretically separates the superior from the inferior parathyroid glands, although not all agree that this relationship exists.²¹ The tubercle is also said to point to the RLN, which will be immediately beneath this vascular fascial layer (Fig. 8).²² On occasion, the tubercle of Zuckerkindl enlarges, from a variety of pathological processes, to form a lobe or an extension that covers the RLN laterally and superficially, enlarging into and occupying the superficial vascular layer. In this circumstance, it is necessary to mobilize the lobe of Zuckerkindl and also rotate it almost 180° anteriorly and medially to expose the underlying RLN. As a consequence, the RLN is often reported to run in a groove posteriorly between the thyroid lobe and the enlarged lobe of Zuckerkindl. The RLN lies in the groove, but it is never actually within the substance of the thyroid.^{23,24}

Two Fascial Layers Surrounding the RLN and the Ligament of Berry

In transverse section at this level, above the inferior thyroid artery, there is a lateral or superficial vascular layer and a medial fibrous, relatively avascular layer (the

ligament of Berry). Once the first vascular layer is dissected and divided, the underlying RLN will be exposed, in turn lying upon the underlying fibrous ligament of Berry, seen in transverse section in Fig. 9.

Viewed from above looking down onto the posterolateral aspect of the left thyroid lobe, after anteromedial rotation after removal of the more superficial anterolateral vascular fascial layer, the RLN is exposed above the level of the inferior thyroid artery, lying upon the ligament of Berry (Fig. 10). The transverse section of the ligament of

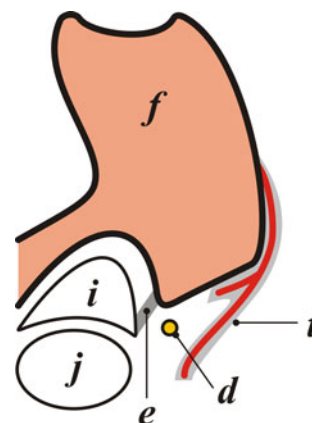


FIG. 9 Two fascial layers surrounding the recurrent laryngeal nerve above the level of the trunk of the inferior thyroid artery, transverse section. Parts as in Fig. 1

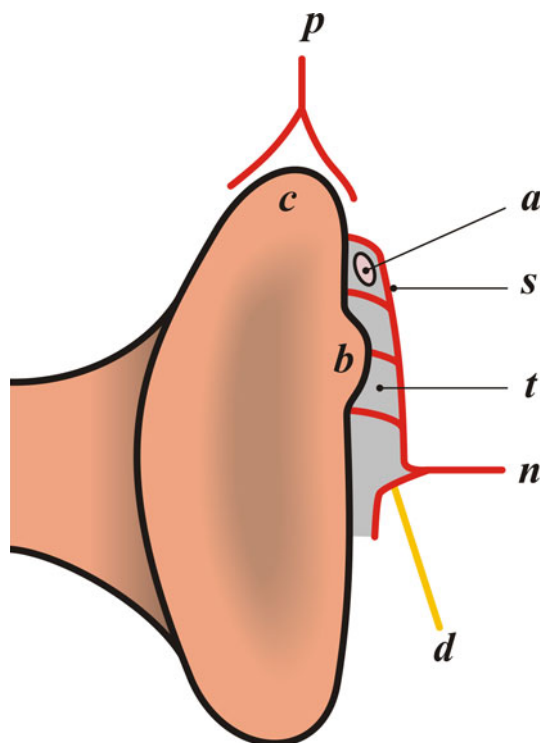


FIG. 8 Tubercle of Zuckerkindl and superior parathyroid gland in the superficial vascular fascial layer. Parts as in Fig. 1

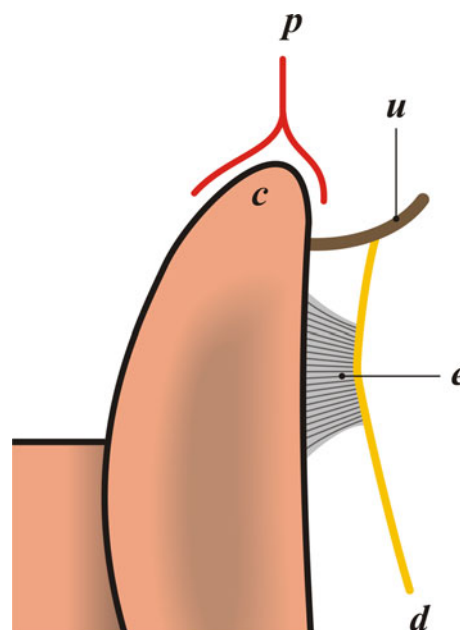


FIG. 10 Left thyroid lobe, rotated medially, surface 2 shaded, after removal of the superficial vascular fascial layer, showing the recurrent laryngeal nerve on lateral aspect of the ligament of Berry, pulled forward into a genu. Parts as in Fig. 1

Berry is shown more simplistically and schematically in Fig. 11.

In the less common setting where the main trunk of the inferior thyroid artery is posterior to the RLN, the anterolateral vascular layer is still present covering the RLN and requires division. However, the main inferior thyroid artery will pass forward at some point behind the RLN and require division. In all other respects, the two fascial layer concept is the same.

The ligament of Berry is now often described as the lateral thyroid ligament.^{9,24} Originally, when describing it in 1888, Berry termed it the suspensory ligament of the thyroid gland.²⁵ This is the more medial and posterior of the two layers described above. I have termed it the true ligament of Berry. In two cadaver anatomical studies, the ligament was found to attach from the posteromedial glandular surface of the thyroid to the inferior margin of the cornu of the cricoid cartilage and inferomedially onto the tracheal wall.^{15,26} Sasou et al. described the ligament of Berry as a band of dense connective tissue extending vertically from the cricoid onto the first to third lateral tracheal rings.¹⁵ The length of the ligament was 8 to 11 mm from trachea to thyroid, and the thickness was 2 to 7 mm (mean, 4.4 mm). The distance of the center of the trachea to the ligament was 10 to 20 mm (mean, 14 mm). Medial to the ligament of Berry is a more loose and vascular connective tissue, connecting the thyroid to the trachea, which can be stripped from the trachea at operation.

At operation, the RLN is thus exposed on the anterolateral surface of the ligament of Berry. However, the thyroid lobe still requires final division from the underlying ligament. The nerve tends to still be attached by loose

areolar fascia to the ligament at this level, and it is thus pulled forward into an artificial genu by the anteromedial location of the thyroid. The nerve is subject to this traction force, which tends to pull it straight, and generates a tensile stress within it, and also a compression force at the site of two fixed points: where the nerve recurs around the aorta or subclavian artery, and where it enters under cricopharyngeus into the larynx.

The ligament of Berry itself often consists of two sheaves, which in turn contain thyroid tissue. The two sheaves with contained thyroid tissue run behind the RLN, and they require removal for complete removal of all thyroid tissue. It is therefore necessary to release the nerve from the underlying ligament of Berry and divide the ligament of Berry. The nerve can be gently pushed laterally and posteriorly while dissecting the underlying ligament of Berry. The ligament of Berry at this level is relatively avascular. However, it often contains an important vessel: the inferior laryngeal artery, which is a posterior branch of the inferior thyroid artery. This vessel requires specific ligation and precise control at this stage. This vessel often passes beneath the RLN. If the vessel is inadvertently divided and retracts, it can be difficult to retrieve behind the RLN and can cause a hematoma at this level. Once the thyroid lobe is released by division of the ligament of Berry, the genu of the RLN disappears, and the nerve tends to fall back into the tracheoesophageal groove, and whereas previously it tended to be straight because of the tension within it, it now tends to become serpiginous, which is itself an indication that dissection of the thyroid lobe and RLN is complete. This process of dividing the ligament of Berry that leads to a dramatic freeing of the thyroid lobe is well known.⁹

However, not all are agreed on the precise anatomy of the ligament of Berry and the relationship of the RLN to it. Lynn report that the RLN runs beneath the ligament of Berry before entering the larynx.¹⁷ Others, including *Gray's Anatomy*, state the RLN may pass laterally, medially, or through the ligament of Berry on occasion.²⁷ Henry states the RLN is embedded in the posterior portion of Berry's ligament.²⁸ Lennquist states the RLN is often invested by the ligament of Berry.²⁹ Wheeler describes the RLN passing through the ligament of Berry.³⁰ Reeve and Thompson also report the RLN passing through the layers of the ligament of Berry.³¹ Hunt described the RLN embedded in the ligament of Berry in 50% of cases.³² I believe these reports represent a terminology and definition problem that may be overcome by the more precise concept of two fascial layers, only the deeper of which represents the true Berry's ligament. Others, in cadaver studies, report that in 96 RLNs dissected, the RLN passes lateral to the true ligament of Berry in all cases.^{15,26}

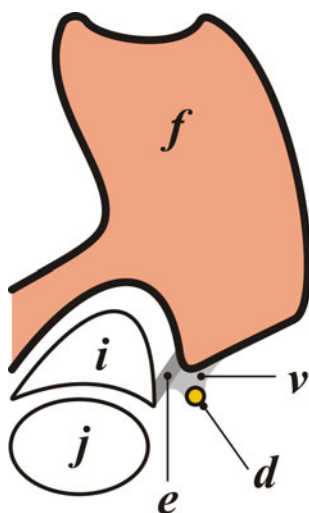


FIG. 11 Transverse section, ligament of Berry, with recurrent laryngeal nerve attached by loose fascial connections. Parts as in Fig. 1

Results of Current Series

That the RLN passes lateral to the true ligament of Berry has also been my experience. In 663 thyroid operations, exposing 977 RLNs over a 69-month period, during which I have studied both bifurcation of the terminal extralaryngeal RLN and the relationship of the RLN to the true ligament of Berry, I observed that all RLNs ran lateral to the ligament.^{33,34}

The overall bifurcation rate was 241 (24.7%) of 977 RLNs. The rate was similar on the right and left sides. The median distance of bifurcation of the RLN to entry under cricopharyngeus on the right and left sides was 18 mm (range, 5–35 mm). In all cases of branched nerves, both the anterior and posterior branches were enveloped between the same two layers as nonbranched nerves: laterally the superficial vascular fascial layer, and medially the true ligament of Berry. Seven patients (1.1%) experienced temporary RLN palsy, but none experienced permanent palsy. There were no known instances of division of anterior or posterior branches individually.

Superior Parathyroid Gland

The superior parathyroid gland is so named because it is situated superior to the inferior thyroid artery, whereas the inferior parathyroid gland is so named because it is inferior to the inferior thyroid artery.

Embryologically, the superior parathyroid gland develops from the epithelium of the dorsal wing of the fourth pharyngeal pouch and attaches to the dorsal surface of the caudally migrating thyroid gland. It does this in association with the ultimobranchial body, which is the ventral component of the fourth pharyngeal pouch, which brings to the thyroid gland the parafollicular cells, which will subsequently produce calcitonin and also marks the site of the tubercle of Zuckerkandl. For this reason, the tubercle of Zuckerkandl is said to point to the superior parathyroid gland. The superior parathyroid gland is more constant in position than the inferior parathyroid gland because of the relatively short descent of the superior gland on the posterior aspect of the descending thyroid lobe. In contradistinction, the inferior parathyroid gland develops from the dorsal wing of the third pharyngeal pouch and has a much longer course, descending with the inferior pole of the thyroid in association with the developing thymus. For this reason, the inferior parathyroid is more variable in position. In 85% of cases, however, the parathyroid glands are grouped at the posterior aspect of the thyroid lobes in an area 2 cm in diameter situated at a point approximately 1 cm above the crossing of the inferior thyroid artery and RLN.

Anatomically, the superior parathyroid is located in the lateral superficial vascular layer of fascia overlying the RLN at a level above the inferior thyroid artery and supplied by the inferior thyroid artery. This is important anatomically when dissecting the superior pole of the thyroid gland so that the blood supply to the superior parathyroid may be recognized and preserved before mobilizing the superior pole. This can be done by mobilizing the fascia of the posterior “v” lip of the superior pole, which will protect, to a degree, the superior parathyroid with its blood supply.

Embryology of the RLN and Superior Laryngeal Nerve

Embryologically, the pharyngeal arches consist of a condensation of mesenchyme, each with its own artery, nerve, muscle, and cartilaginous elements. The fourth pharyngeal arch develops the cricothyroid muscle and has as its nerve the superior laryngeal nerve. This gives a branch, the external laryngeal nerve, that supplies the cricothyroid. The sixth pharyngeal arch develops all other intrinsic muscles of the larynx and its nerve is the RLN or the nerve of the sixth pharyngeal arch. Both the superior laryngeal nerve and the RLN are branches of the vagus (10th cranial) nerve.

Variation in the Anatomy of the RLN

Extralaryngeal bifurcation of the RLN will occur in approximately 24% of cases, although rates of up to 70% have been reported by some. The rates for bifurcation are similar on the right and left side. However, when bifurcation occurs on the left side, it predicts bifurcation on the right in 43% of RLNs.³³ The median distance from the point of bifurcation of the RLN to entry under the cricopharyngeus is of the order of 18 mm on the right and left sides. A recent study has shown that the motor fibers to both the adductors and abductors of the larynx are located in the anterior extralaryngeal branch of the RLN and that no motor fibers are located in the posterior branch.³³ Therefore, damage to the anterior branch is particularly important, if it is believed the sole RLN is the posterior division of the RLN. In this situation, the anterior branch may be inadvertently damaged while the operator believes the whole nerve has been preserved, yet all of the motor fibers will have been divided. Further, when the RLN is bifid, the anterior extralaryngeal branch will be smaller in diameter than the undivided trunk of the RLN. In this situation, for a given degree of traction, the tensile force generated within the smaller anterior branch will be greater than in the nerve trunk, and hence traction injury and neurapraxia are more likely in a bifid RLN. These two

observations may explain why RLN palsy is more common when the nerve is bifid.³⁵

Nonrecurrence of the right RLN will occur in 0.5% of cases. Nonrecurrence does not occur on the left side.³⁶ This is a consequence of the different embryology on the two sides. The anomaly only occurs when there is an anatomical abnormality of the right subclavian artery. Non-RLNs are at greater risk of injury laterally in the neck; however, they tend to pass anteromedially either at or just below the level the inferior thyroid artery and then ascend in the normal position within the tracheoesophageal groove. Therefore, above the inferior thyroid artery, they are probably not at increased risk.

CONCLUSIONS

Awareness of the two fascial layers enveloping the last 2 cm of the extralaryngeal course of the RLN enables safe division of the superficial vascular fascial layer, fully exposing the underlying branched or nonbranched RLN lying laterally upon the true ligament of Berry. This region will usually be the last part of the thyroid lobe to be dissected from surrounding tissues. Care not to exert too much anteromedial traction will help to minimize tension in the RLN and reduce the degree of artificial genu formation of the nerve. Gentle posterior dissection of the RLN posterolaterally through the loose areolar tissue attaching it to Berry's ligament will also aid reduction of tension within the nerve. Safe division of the true ligament of Berry is then possible, allowing the thyroid lobe to spring forward and the RLN to drop back, adopting a serpiginous appearance indicating complete dissection of the thyroid lobe. This knowledge and these maneuvers will potentially help reduce RLN palsy at this site, where it is at greatest risk.

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